Twitch Sponsorship Optimization Agent

Outcomes Report: Initial and Updated Problem Framing

MGMT 590-037 | AI-Enhanced Optimization | Daniels School of Business, Purdue University

Team 1: Ethin DalPian Nick Sopcic Matt Foor Shubh Vaishnav

Summer 2026

Executive Summary

We built an end-to-end AI agent that takes a sponsorship budget, fits audience response curves to Twitch session data, and returns an optimal per-segment allocation with a plain-language justification. Across both the initial framing (V1) and the stakeholder-modified framing (V2), the agent consistently outperforms intuitive heuristics.

Key findings:

- At a \$20,000 budget the optimized agent produces **8.97 incremental conversions**, a 2.3% gain over historical allocation and a 630% gain over a reach-chasing baseline, both statistically significant (bootstrap 95% CI lower bound > 0).
- The reach-chasing heuristic — spend on the largest-follower segment first — is the worst-performing strategy because the largest segment (Mega) has a poorly identified response curve and delivers near-zero measured lift.
- When per-segment conversion quality is incorporated (V2 stakeholder values: \$18–\$31 per conversion by tier), the revenue metric rises from \$23.53 to **\$249.99** at the same allocation, and the agent's revenue advantage over reach-chasing grows to \$222.94 (median bootstrap uplift).
- A data-fusion estimate of per-segment values produced a near-flat vector (\$0.82–\$1.24) because the k-means clusters have similar session characteristics. The stakeholder-supplied tier values carry more information and are recommended for operational use.

Primary recommendation: deploy the agent with Model B (stakeholder per-segment values) and a budget of \$15,000–\$20,000, targeting Segments 1 (Mid, \$31/conversion) and 3 (Micro, \$27/conversion). Expand the creator pool in Segment 3 before increasing that segment's budget, as its spend cap is already binding.

1 Problem Framing (V1 — Initial)

1.1 Business Problem

A gaming studio wishes to allocate a fixed sponsorship budget across Twitch creator segments to maximize incremental in-game conversions. Conversions are game actions (purchases, sign-ups) that occur within three hours of a sponsored session. The key challenge is that the relationship between sponsorship spend and conversions is non-linear: audience response follows an S-shaped Hill curve with increasing returns at low spend, an inflection point, and diminishing returns at high spend. The resulting optimization is a non-convex NLP, requiring a global search method rather than a standard gradient solver.

1.2 Data and Segmentation

The pipeline ingests 39,928 Twitch sessions across 2,000 creators and 12,000 player records. Creators are grouped into four segments using k-means on log-transformed creator characteristics (followers, unique viewers, session duration, sponsorship rate). The player dataset provides calibration- period behavioral features used to estimate a per-conversion dollar value.

Table 1. Segment profile and fitted Hill curve parameters

Seg.	Tier	Med. Followers	V	K (\$)	n	Spend Cap (\$)
0	Macro	99,336	1.23	6.01	3.36	6,173
1	Mid	96,419	3.49	3.05	3.00	5,978
2	Mega	125,980	448.73	33,888	20.00†	14,612
3	Micro	66,074	4.41	376.30	1.41	3,846

†Segment 2 Hill fit hit the upper bound ($n = 20$). Parameters are poorly identified; results for this segment should be treated as approximate.

1.3 Optimization Formulation

The agent solves a non-convex NLP:

$$\max_{\mathbf{x}} \sum_s g_s(x_s) \quad \text{s.t.} \quad \sum_s x_s \le B,\ 0 \le x_s \le u_s$$

where $g_s(x_s) = V_s x_s^{n_s} / (K_s^{n_s} + x_s^{n_s})$ is the Hill response curve for segment s. With $n_s > 1$, each curve is S-shaped, producing a combined objective surface with genuine local optima when budget is split across segments in their convex regions. Basin-hopping with SLSQP as the local minimizer is the primary solver; dual annealing serves as validation. Both converged to identical objectives (gap $< 10^{-6}$), providing strong evidence the global optimum was found.

The CVXPY library rejected the Hill objective as non-DCP, confirming non-convexity. A single SLSQP run from a low-spend start was trapped at a local boundary maximum (f = 111.7), while basin-hopping escaped to the global optimum (f = 170.0), a 52% improvement on the benchmark problem.

2 V1 Results — Initial Problem Framing

2.1 Prediction Layer

The prediction pipeline (Component 2) estimates a per-session incremental lift $\hat{\tau}$ using a T-learner: two separate XGBoost models, one trained on sponsored sessions and one on unsponsored. The T-learner was selected over the S-learner because it allows each treatment arm to capture arm-specific feature interactions. Lift estimates from an IPW-reweighted S-learner confirmed stability (pairwise correlation ≥ 0.83 across all three estimators).

Hill saturation curves are then fitted to $(sp\_cost, \hat{\tau})$ pairs per segment via `scipy.optimize.curve_fit`. A player value model (two-stage XGBoost: classifier predicting spender status, regressor predicting spend amount) yields a flat per-conversion value $v = \$2.623$.

2.2 Optimal Allocation

Table 2. V1 optimal allocation at budget \$20,000 (flat $v = \$2.623$)

Seg.	Tier	Spend ($)	Incr. Conversions	Incr. Revenue ($)	Funded
0	Macro	5,000.00	1.2295	3.23	Yes
1	Mid	5,000.00	3.4885	9.15	Yes
2	Mega	5,000.00	0.0000	0.00	Yes†
3	Micro	3,846.00	4.2517	11.15	Yes
Total		**18,846.00**	**8.9697**	**23.53**	

†Segment 2 lift is effectively zero due to the poorly identified Hill fit ($K = \$33{,}888$, cap $= \$14{,}612$).

Segment 3 hits its spend cap (\$3,846), the only binding constraint. Segment 2 receives \$5,000 but delivers zero measurable lift, representing an inefficient allocation driven by the unreliable Hill parameters.

2.3 Baseline Comparison and Validation

Table 3. Strategy comparison on held-out test creators (300 creators)

Strategy	Spend ($)	Incr. Conversions	Incr. Revenue ($)
Agent (Optimized)	18,846.00	8.9697	23.53
Historical	18,577.01	8.7711	23.01
Reach-Chasing	20,000.00	1.2295	3.23
Agent vs. Historical (%)		**+2.3%**	**+2.3%**
Agent vs. Reach-Chasing (%)		**+629.9%**	**+629.3%**

Table 4. Bootstrap uplift (1,000 resamples of held-out creators)

Comparison	Median Uplift	95% CI Lo	95% CI Hi	Win?
vs. Historical	0.1992	0.0959	0.3820	**Yes**
vs. Reach-Chasing	7.7402	7.7402	7.7402	**Yes**

Win declared when 95% CI lower bound > 0.

The agent wins against both baselines. The narrow confidence interval against reach-chasing reflects deterministic segment follower rankings; regardless of which creators appear in the bootstrap sample, the agent's avoidance of the zero-lift Mega segment always produces a large advantage. The tighter margin over the historical baseline is expected: the historical allocation already approximates the optimum because it reflects practitioners' implicit knowledge of which segments convert.

Sensitivity highlights. The objective is most sensitive to Segment 3's saturation parameter V (max $|\Delta\text{obj}| = 0.425$ per $\pm 10\%$ perturbation). Segment 3's cap is binding; relaxing it by 50% yields $+0.068$ additional conversions.

3 Updated Problem Framing (V2 — Stakeholder Modification)

3.1 Motivation

The stakeholder identified that the flat value weight $v = \$2.623$ assigned equal revenue importance to every conversion regardless of which creator segment delivered it. Audience quality varies across tiers: niche, loyal micro-creator communities tend to convert at higher lifetime value than mega-creator audiences, who skew toward passive viewership. The modification introduces a per-segment value vector $\{v_s\}$ and re-solves the allocation under three value models.

3.2 Phase 1 — Stakeholder-Supplied Values

Segments are ranked by median creator follower count and assigned to the four stakeholder-defined tiers. The Mega tier — despite the largest raw audience — receives the lowest per-conversion value ($18), reflecting lower audience spend propensity.

Table 5. Phase 1 segment-to-tier mapping and stakeholder values

Seg.	Tier	Med. Followers	Rank	v_s – Stakeholder ($/conv.)
2	Mega	125,980	1 (highest)	$18.00
0	Macro	99,336	2	$22.00
1	Mid	96,419	3	$31.00
3	Micro	66,074	4 (lowest)	$27.00

3.3 Phase 2 — Data Fusion Estimate

Because no shared key links the player and Twitch datasets, v_s cannot be estimated directly. We apply a bridging approach using **platform engagement depth** as the shared construct: `pmins` (total gameplay minutes) on the player side and `duration` (session length in minutes) on the Twitch side. Both variables measure willingness to invest time in the platform; neither is a follower-count or audience-size proxy.

The procedure ranks both variables on a $[0, 1]$ percentile scale, fits a binned mean $\hat{v}(\text{pmins\_percentile})$ on the player data, then evaluates this function at the mean duration-percentile of each Twitch segment.

Table 6. Three-model value weight comparison

Value Model	Seg 0 Macro	Seg 1 Mid	Seg 2 Mega	Seg 3 Micro
(A) Flat v	$2.62	$2.62	$2.62	$2.62
(B) Stakeholder v_s	$22.00	$31.00	$18.00	$27.00
(C) Fusion v_s	$0.82	$1.24	$1.04	$0.95

Fusion limitation. The four k-means segments have similar mean session durations (130–142 minutes), so the duration bridge produces a near-flat fusion vector (range: $0.43). A sensitivity check using a second bridge (`chat_rate_pm` / viewer retention ratio) produced an equally flat result. This confirms that the k-means segmentation captures engagement quality dimensions that the available creator-level covariates do not differentiate. Model B (stakeholder values) carries materially more information.

3.4 Three-Way Optimization Results

The objective with per-segment values is $\max_{\mathbf{x}} \sum_s v_s\, g_s(x_s)$. All three models were solved with basin-hopping ($n_{\text{iter}} = 200$) and validated against dual annealing.

Table 7. Three-way comparison at budget $20,000

Model	Seg 0 ($)	Seg 1 ($)	Seg 2 ($)	Seg 3 ($)	Incr. Conv.	Revenue ($)
(A) Flat	3,870	2,194	4,293	3,846	8.9697	23.53
(B) Stakeholder	3,870	2,194	4,293	3,846	8.9697	249.99
(C) Fusion	3,870	2,194	4,293	3,846	8.9697	9.39

4 V1 vs. V2: Analysis and Recommendations

4.1 V1 vs. V2 Comparison

Table 8. Summary comparison: V1 initial framing vs. V2 updated framing

Dimension	V1 — Initial Framing	V2 — Updated Framing
Value objective	Flat scalar $v = \$2.623$ applied uniformly to all segments	Per-segment vector $\{v_s\}$; three competing models (A/B/C)
Optimization obj.	$\max \sum_s g_s(x_s)$ (pure conversions)	$\max \sum_s v_s g_s(x_s)$ (revenue-weighted)
Allocation found	Seg 0–2: $5,000 each; Seg 3: $3,846	Seg 0: $3,870; Seg 1: $2,194; Seg 2: $4,293; Seg 3: $3,846
Revenue metric	$23.53 (flat v)	$249.99 (Model B stakeholder v_s)
Bootstrap uplift	+0.199 conv. vs. Historical (WIN); +7.74 vs. Reach-chasing (WIN)	+$222.94 revenue vs. Reach-chasing for Model B (WIN)
Key limitation	Hill fit for Seg 2 ($n = 20$) unreliable; Seg 2 assigned $5,000 with zero lift	Fusion estimate near-flat; stakeholder values preferred

4.2 Key Discussion Points

Q1: Are the flat-v and pure-conversions allocations identical? Yes, mathematically and numerically. For any positive scalar v, $\arg\max_{\mathbf{x}} v \sum_s g_s(x_s) = \arg\max_{\mathbf{x}} \sum_s g_s(x_s)$. The two-norm difference between the V1 allocation (pure conversions) and Model A (flat-v revenue) is $0.00 to four decimal places. Model A therefore represents both objectives simultaneously.

Q2: How do differentiated v_s values move spend? At the tested budget ($20,000), the physical allocation is unchanged across all three models. This is a direct consequence of the near-saturated Hill curves: with K values of $3–$376 for Segments 0, 1, and 3, all feasible spend levels already operate in the diminishing-returns region, and the optimizer cannot productively redirect capital because every segment's marginal lift is similar. The revenue metric changes dramatically (from $23.53 to $249.99) because the v_s multipliers are 7–12× larger, not because the allocation changed.

Q3: Do Models B and C point in the same direction? Broadly yes: both assign the lowest weight to Segment 2 (Mega) and the highest to Segment 1 (Mid). However, Model C's range ($0.43) is negligible compared to Model B's range ($13), and both converge to identical allocations in practice. Model B is preferred because it encodes explicit audience-quality knowledge derived from domain expertise. Model C demonstrates the fusion methodology but cannot recover meaningful differentiation from the available creator-side covariates.

4.3 Recommendations

1. **Deploy with Model B (stakeholder values).** The revenue metric under stakeholder v_s is the most informative objective. Use budget $15,000–$20,000; the agent's advantage over reach-chasing is robust

across this range (bootstrap lower bound > \$222).

2. **Expand the Segment 3 creator pool.** Segment 3 (Micro, $v_s = \$27$, second-highest value) is the only binding cap constraint. Adding creators to this tier would allow the optimizer to allocate more capital to the most revenue-efficient audience without reducing spend elsewhere.
3. **Re-fit the Segment 2 Hill curve with additional data.** The current fit is unreliable ($n = 20$, at the upper bound). The optimizer correctly assigns zero effective lift to Segment 2, but this means \$4,293–\$5,000 of each run's budget is either wasted or left unspent. Collecting more sponsored sessions with varied spend levels in Segment 2 would allow a stable fit and potentially unlock material lift from the largest-audience tier.
4. **Validate the lifetime-value assumption.** The flat value $v = \$2.623$ reflects first-transaction revenue only. Multiplying by a retention-adjusted lifetime-value factor (the agent stores $v_{\text{adj}} = \$1.03$ using observed churn rates) would reduce the absolute revenue estimates but not change the optimal allocation. Before budget decisions are made, the team should confirm which value definition the business uses for ROI calculations.
5. **Accept the sponsorship-causality limitation.** Sponsorship is not randomized in this dataset. The T-learner and IPW robustness check control for observed confounders, but residual unmeasured confounding (e.g., seasonal creator popularity) means the lift estimates are observational, not causal. The agent's advantage over historical allocation is small (+2.3%) precisely because the historical pattern already approximates a reasonable allocation. The large advantage over reach-chasing (>600%) reflects a structural insight—audience quality matters more than audience size—that is robust to confounding.

4.4 Conclusion

The Twitch Sponsorship Optimization Agent demonstrates that a four-stage pipeline (data, prediction, optimization, explanation) can convert raw session logs into a defensible, revenue-aware sponsorship strategy. The core insight running through both the V1 and V2 analyses is that the Mega-follower segment is a trap: its large audience does not translate into measurable incremental lift or high-quality conversions. Mid-tier and micro-niche creators deliver more conversion value per dollar. Introducing per-segment value weights (V2) sharpens this finding quantitatively—the revenue gap between the agent and the reach-chasing heuristic grows from 630% (conversions) to nearly 10× (revenue) under stakeholder values—and provides the business with an actionable tier-differentiated budgeting framework.

AI use disclosure: This project utilized Claude (Anthropic) for code generation, notebook scaffolding, and report drafting. All model outputs, data interpretations, and recommendations were reviewed and validated by the team.